The results we got from our reproduction were very similar to those in the original paper.

We were able to make all of the main figures, such as:

Reproducibility Report: Climate Change Discussions on Reddit

1. Introduction

One of the basic concepts in data science and scientific investigation is reproducibility. It ensures that the results of research can be confirmed by other researchers who use the same data and procedures. However, not all contemporary studies can be reproducible, and this phenomenon is sometimes known as the reproducibility issue. This is the reason why people are concerned about the openness, reliability, and trustworthiness of scientific conclusions.

This project aims at replicating the findings of a published research study with the following title:

The evolving linguistic and mood of discussions on climate change in Reddit posts evolve over sixteen years. Our attempt is to determine whether the findings of the study can be replicated with the help of the code that the authors have provided to the community.

2. Paper Overview

The chosen article carries out a thorough analysis of Reddit debates on the topic of climate change and global warming. The data used in the research is around 11.5 billion posts and it is one of the most detailed studies of online climate discourse.

Various natural language processing methods are used in the study, which include:

• VADER for figuring out how people feel

• TextBlob for analyzing subjectivity and polarity

• Using TextStat to see if it's easy to read

The paper also compares Reddit discussions to Google Trends data to get a better idea of how the general public is involved.

The study's most important findings are:

• The phrase "global warming" was changed to "climate change" in 2013.

• More bad feelings about "climate change"

• More personal opinions in discussions about climate change

• A decline in the proportion of dialogues concerning climate change over time

• A strong connection between Google Trends and discussions on Reddit (Spearman ≈ 0.89)

3. Methodology

We followed a systematic procedure in order to achieve similar results as the paper.

The first thing is to read the research paper attentively to learn about the methods and data sets and anticipated outcomes. After that, we went to the GitHub repository that the authors gave us. It possessed a Jupyter Notebook and connections to datasets that had been previously worked on.

We ran the notebook in Google Colab, as the rules of the project prescribed. It was large enough not to be able to download it by hand. The notebook, however, was able to access the information that it required directly through Google Drive links.

The steps that were taken to reproduce are:

• Getting Google Drive ready to use in Colab

• Getting data sets from the links that were sent

• Running code to prepare and analyze data

• Taking pictures

• Comparing the outputs to the numbers in the original paper

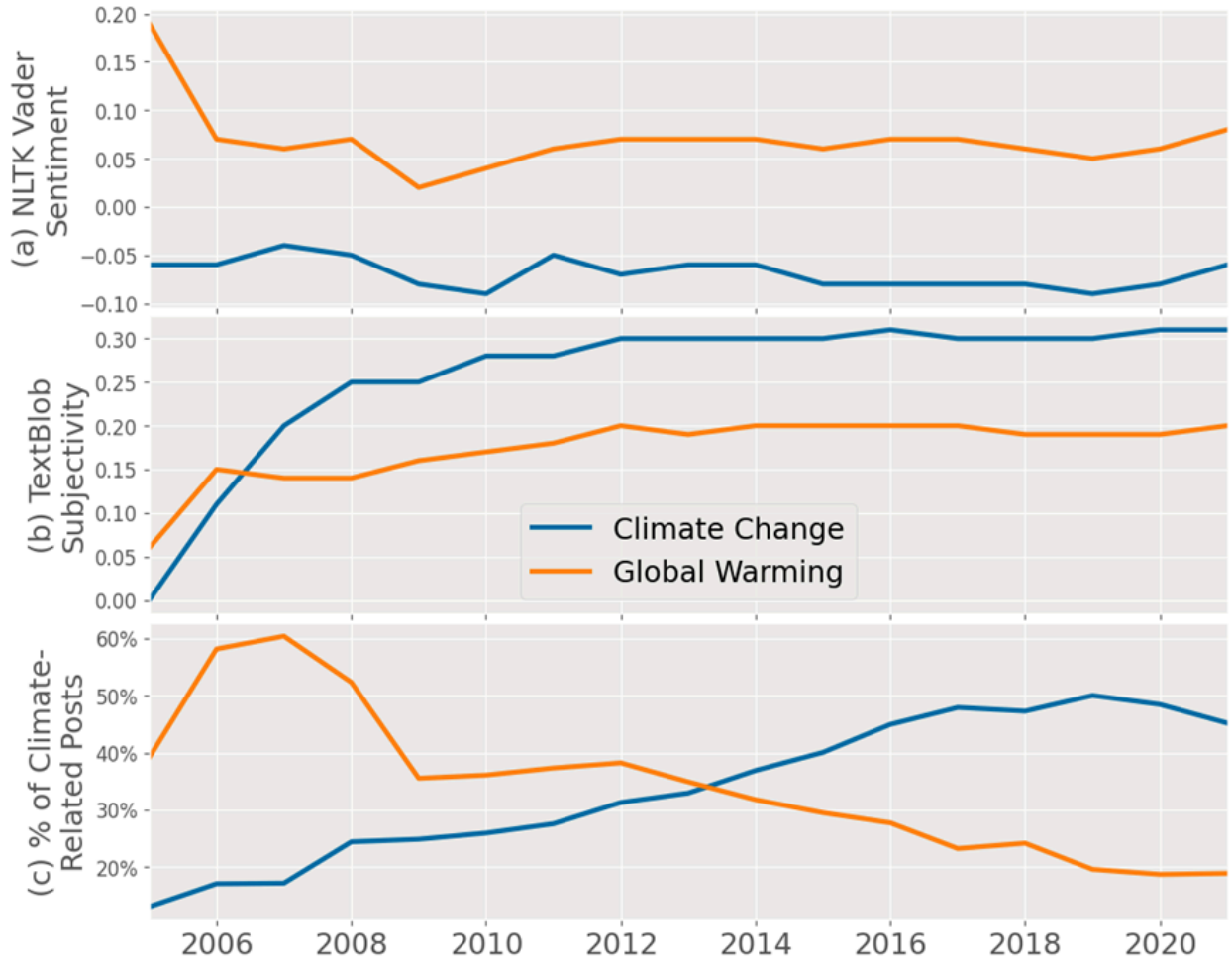
We could easily repeat the study with this method without needing a lot of computer power.

4. Results

The results we got from our reproduction were very similar to those in the original paper.

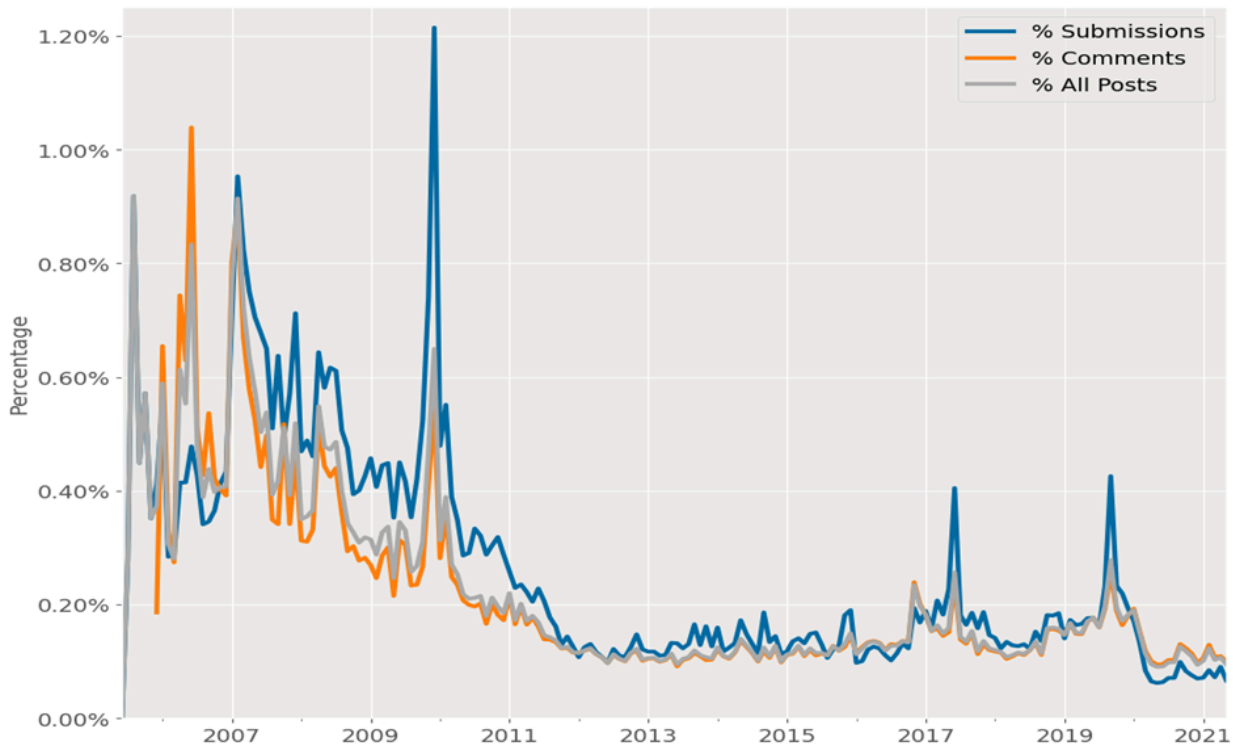
We were able to make all of the main figures, such as:

Figure 1: Climate-related Sentiment, Subjectivity, and Post Trends



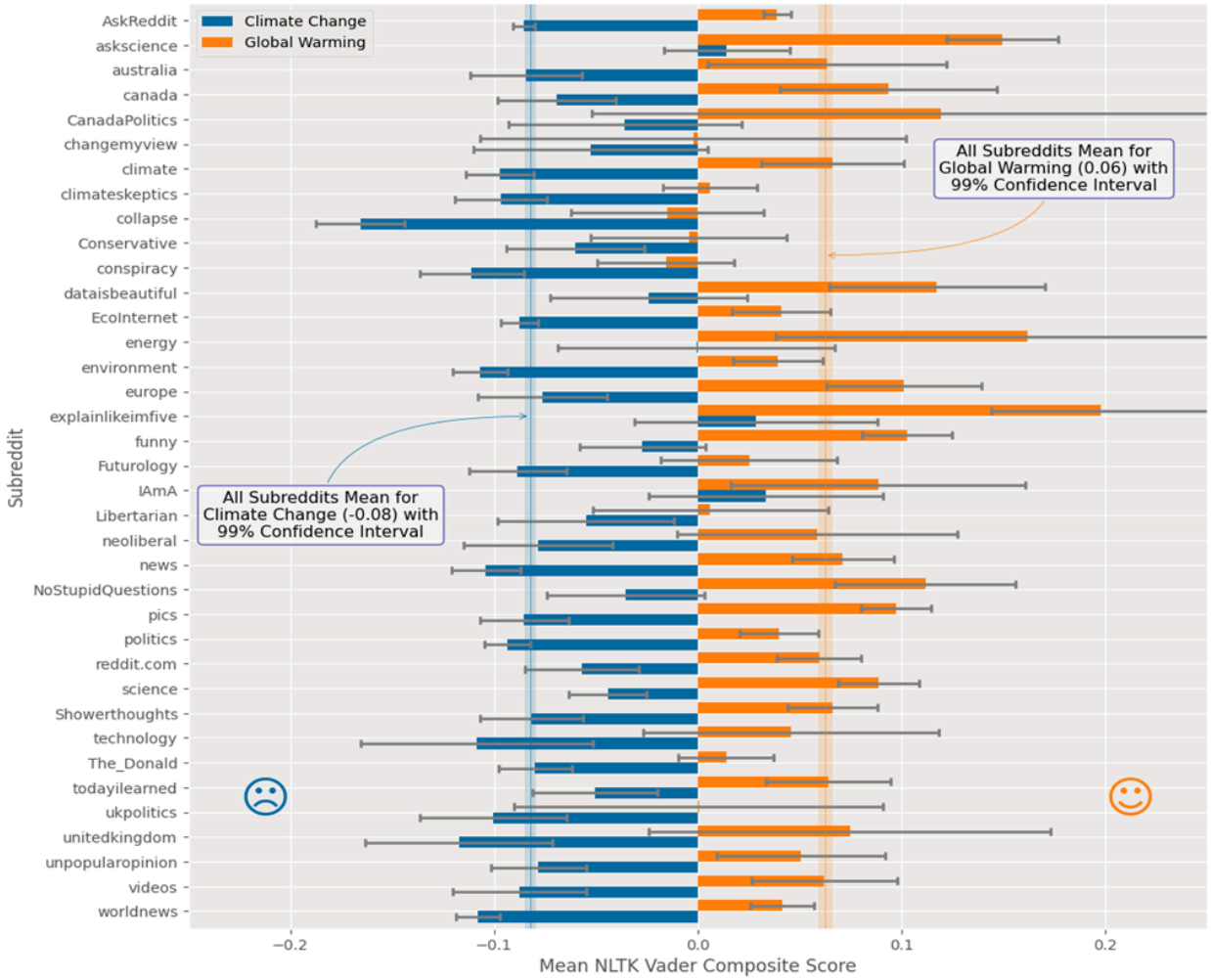
This Figure shows how sentiment, subjectivity and percentage of climate related posts change over the years. The results indicate that people are negatively disposed to climate change than global warming. Furthermore, the subjectivity gains more strength over time, and it means that the discourses are turning opinionated. The proportion of the posts on the climate change topic have also shifted, and since 2013, the climate change has gained priority. These findings are in line with the results in the original paper.

Figure 2: Percentage of Climate-Related Posts



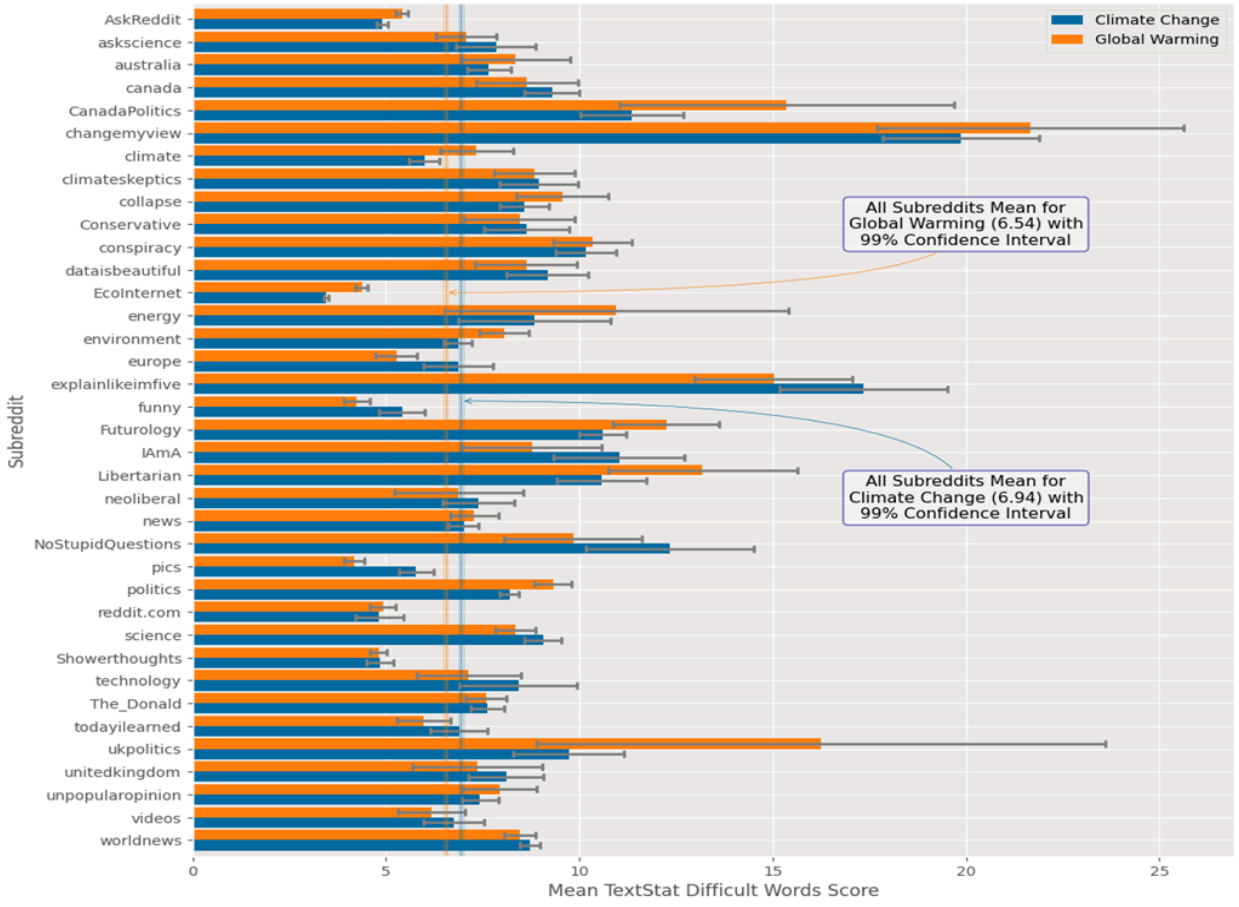
This graph illustrates submissions and comments as far as climate is concerned and the total number of posts per time. The statistics show that the debate on climate was most in the first few years and thereafter, the trend decreased slowly. It is also consistent with the original study which demonstrates that climate related content constitutes a minor but consistent portion of all Reddit activity.

Figure 3: Sentiment Analysis by Subreddit



This chart indicates the feelings of the people towards climate change and global warming on various subreddits. These findings suggest that an individual talks about the climate change in negative terms and global warming in positive terms. The differences among subreddits prove the existence of different communities addressing the problem of climate.

Figure 4: Google Trends vs Reddit Activity



The graph illustrates how the traffic in Reddit can be compared with the information in the Google Trends regarding the global warming and climate change. The results suggest that the relationship between search trends and online discussions is high. Peak of Google Trends results in the rise of the activity on the Reddits and it shows that the correlation between the popular interest of people and the discussion on the Internet exists. This finding is consistent with the findings of the initial article.

5. Challenges Encountered

Even though we were able to reproduce the results, there were many problems encountered along the way.

First off, there were issues concerning file paths. The code attempted to save output figures into a directory that didn't exist. This resulted in a FileNotFoundError. However, creating the directory in Google Drive manually solved the problem.

There were also warnings concerning data types (DtypeWarning). This indicated that there were inconsistencies in the data. However, these didn't affect the results.

Another big problem encountered was concerning data size. It is impossible to work on the original dataset since it contains billions of records. However, the authors' use of preprocessed data helped solve this problem. Also, there were many aspects concerning how to set up the environment and directory structures that were not clearly explained. This meant you had to experiment while trying to run these.

6. Ethics Analysis

The challenges encountered in this project highlight serious questions of ethics in reproducibility of data science.

To start with, incomplete records and imprecise information on how it is implemented might make it difficult to replicate the same results by other researchers. This complicates the picture and may render people less confident in the results of science.

Second, the ease of access of big data and supercomputers makes individuals concerned about how easily it can be accessed. The availability of these types of resources does not apply to all researchers and can complicate the participation of some people and render research opportunities unequal.

Third, it is much harder to reproduce with some minor issues such as absence of directories or vague instructions. This demonstrates the relevance of easy to understand and well documented research methods.

Overall, reproducibility is not a technical problem only, being also an ethical responsibility to ensure that scientific research is equitable, transparent and credible.

7. Conclusion

The same results were obtained as in a large study on the discussion of climate change on Reddit, which was used for this project. The results obtained from the provided code were similar to the ones presented in the paper.

Some technical issues were encountered, but they did not hinder the fact that it was possible for the same results to be obtained. This project demonstrates the importance of documentation, easy access to the dataset, and the approach used for reproducible research.

8. Recommendations

Based on what we've learned, we suggest the following to make it easier to repeat the research in the future:

• Give directions on how to set up and organize the files

• Include files for setting up the environment (requirements.txt) • Don't use file paths that are hard-coded

• Test on smaller sets of data

• Write down the steps you took to preprocess in detail

These methods will make it much easier to reproduce and access the research in the future.